NAIL CLUBBING

A Comprehensive Review for Physiotherapy Practice

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INTRODUCTION & DEFINITION

Definition

Digital clubbing (also known as Hippocratic fingers) is a clinical sign characterised by bulbous, uniform swelling of the soft tissue of the terminal phalanx of a digit, with subsequent loss of the normal angle between the nail and the nail bed.

KEY FEATURES

- Increased nail curvature in both longitudinal and transverse axes
- Sponginess (fluctuation) of the nail bed on palpation
- Loss of the normal 160° Lovibond angle (angle becomes ≥ 180°)
- Increased distal phalangeal depth (DPD) to interphalangeal depth (IPD) ratio > 1
- Can be unilateral or bilateral; may be hereditary or acquired

HISTORICAL BACKGROUND

> "In persons affected with empyema … the nails of the hands are bent, and the fingers become warm, especially at their tips."
>
> — Hippocrates (c. 460–370 BCE), Prognostics

~400 BCE	First described by Hippocrates in patients with empyema — hence also called 'Hippocratic fingers'
19th Century	Systematic clinical categorisation began; association with pulmonary diseases established
1938	Lovibond described the profile angle measurement for objective grading
1976	Schamroth described his 'window test' as a simple bedside screening method
Modern Era	VEGF and platelet-related theories provide better understanding of pathogenesis

PATHOPHYSIOLOGY & MECHANISM

The exact mechanism is not fully understood, but several theories have been proposed. The most widely accepted involves VEGF and megakaryocyte/platelet fragments.

VEGF / Hypoxia Theory

- Chronic hypoxia → ↑ VEGF release from distal tissues
- VEGF promotes angiogenesis & vascular permeability
- Connective tissue hyperplasia in the nail bed
- Leads to soft-tissue swelling of the fingertip

Megakaryocyte / Platelet Theory

- Megakaryocyte fragments normally fragmented in lungs
- In lung/cardiac disease → fragments bypass the lungs
- Lodge in distal capillaries of the digits
- Release PDGF & VEGF → local tissue proliferation

Neural & Genetic Factors

- Vagal-mediated mechanism (clubbing resolves after vagotomy in some cases)
- PGE_2 overproduction implicated in primary hypertrophic osteoarthropathy
- Hereditary / idiopathic forms (pachydermoperiostosis) linked to HPGD gene
- Unilateral clubbing suggests local vascular aetiology

ETIOLOGY — CARDIOPULMONARY CAUSES

PULMONARY CAUSES

- Bronchogenic carcinoma (most common malignant cause)
- Bronchiectasis
- Cystic fibrosis
- Idiopathic pulmonary fibrosis (IPF)
- Lung abscess
- Mesothelioma
- Chronic empyema
- Pulmonary arteriovenous malformations

CARDIAC CAUSES

- Cyanotic congenital heart disease (Tetralogy of Fallot, etc.)
- Infective endocarditis (subacute)
- Atrial myxoma
- Arteriovenous fistulae
- Aortic aneurysm (secondary infection)

ETIOLOGY — OTHER SYSTEMIC CAUSES

Gastrointestinal

- Inflammatory bowel disease (Crohn's, UC)
- Hepatic cirrhosis
- Coeliac disease
- GI lymphoma

Endocrine

- Graves' disease (thyroid acropachy)
- Hyperparathyroidism

Hereditary / Idiopathic

- Primary hypertrophic osteoarthropathy (pachydermoperiostosis)
- Familial / idiopathic clubbing (no underlying disease)

Other

- Unilateral: AV fistula, axillary artery aneurysm, Pancoast tumour
- Pregnancy-related (resolves post-partum)
- Chronic infections (TB, HIV)

GRADES OF CLUBBING

Grade 1

Fluctuation & Softening

- Softness and fluctuation of the nail bed on palpation
- Loss of normal firm attachment
- Nail bed becomes spongy when pressed
- Lovibond angle: ~160° (normal variant)

Grade 2

Loss of Lovibond Angle

- Obliteration of the normal Lovibond angle
- Angle between nail plate and proximal nail fold ≥ 180°
- The profile ratio (DPD/IPD) exceeds 1.0
- Nail appears to 'float' on a soft tissue bed

Grade 3

Increased Curvature

- Accentuated convex curvature of the nail in all planes
- Both longitudinal and transverse curvature increased
- Parrot-beak appearance when viewed from side
- Clear bulbous enlargement of the fingertip

Grade 4

Drumstick Appearance

- Entire distal phalanx swollen — 'drumstick' shape
- Often associated with hypertrophic osteoarthropathy (HOA)
- Wrist & ankle periostitis may coexist
- Represents most severe form of clubbing

CLINICAL ASSESSMENT & TESTS

1. Schamroth Window Test

Place dorsal surfaces of opposite terminal phalanges together. Normally a diamond-shaped window is visible. In clubbing, this window is obliterated.

Sensitivity: ~93% | Specificity: ~95% | Quick bedside screening tool

2. Lovibond Angle / Profile Sign

The angle between the nail plate and the proximal nail fold (hyponychial angle) is measured. Normal: < 180° (approximately 160°). Clubbing: ≥ 180°.

Objective measurement | Can be quantified with a shadowgraph or digital photography

3. Phalangeal Depth Ratio (PDR)

Ratio of the Distal Phalangeal Depth (DPD) to the Interphalangeal Depth (IPD). Normal: DPD/IPD < 1.0. Clubbing: DPD/IPD ≥ 1.0.

Most objective and reproducible clinical measurement of clubbing

4. Nail Bed Fluctuation Test

Press on the nail bed — in early clubbing, the nail feels 'floating' or spongy due to underlying soft tissue oedema and increased vascularity.

Earliest sign of clubbing | Highly subjective but clinically useful

SCHAMROTH WINDOW TEST — DETAIL

✓ NORMAL FINDING

When opposing terminal phalanges of the same finger are placed together (nail-to-nail), a small diamond-shaped 'window' of light is visible at the nail base.

This window exists because the Lovibond angle is < 180°, creating a gap between the two nail-fold junctions.

The test was developed by Leo Schamroth in 1976 after he noticed the disappearance of this window during his own episode of infective endocarditis.

✗ POSITIVE (CLUBBING)

When opposing phalanges are placed together, the normal diamond-shaped window is ABSENT (obliterated).

This occurs because: the Lovibond angle ≥ 180° → the nail-fold areas are flush against each other → no gap remains.

Interpretation:

- Window present → No clubbing (negative)
- Window absent → Clubbing present (positive)

Limitation: Less reliable in early/grade 1 clubbing; combine with phalangeal depth ratio for better accuracy.

DIFFERENTIAL DIAGNOSIS

Not everything that looks like clubbing IS clubbing. Key differentials to consider:

Pseudoclubbing

Nail curvature increased but NO increase in soft tissue. Seen in hyperparathyroidism, renal osteodystrophy. DPD/IPD ratio remains < 1.0.

Koilonychia

Spoon-shaped (concave) nails — opposite of clubbing curvature. Associated with iron-deficiency anaemia.

Paronychia

Infection of the periungual tissue causing localised swelling. Painful & usually unilateral / single digit.

Felons / Whitlows

Swelling of the fingertip pulp due to infection. Tender, warm, and erythematous — distinct from painless clubbing.

Onycholysis

Separation of nail plate from nail bed. Can mimic the 'floating nail' appearance but has different aetiology.

RELEVANCE TO PHYSIOTHERAPY

Clinical Assessment

- Clubbing is a red flag: may indicate underlying cardiopulmonary pathology
- Part of general hand inspection in every cardiorespiratory examination
- Progressive clubbing in a patient → refer for medical investigation
- Regression of clubbing may indicate response to treatment (e.g. post-surgery)

Cardiopulmonary Rehabilitation

- Patients with clubbing often have chronic lung disease (COPD, IPF, bronchiectasis)
- Tailor exercise prescription based on underlying diagnosis
- Monitor SpO_2 — hypoxia may be present even at rest
- Airway clearance techniques crucial in bronchiectasis / CF patients

Patient Education

- Explain the significance of clubbing as a systemic sign
- Motivate adherence to pulmonary rehab programmes
- Teach self-monitoring (oxygen saturation, symptom diaries)
- Encourage smoking cessation in relevant cases

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